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$$\begin{aligned} & \int \frac{dx}{x^2 + 5} \\ &= \int \frac{dx}{5 \left(\frac{x^2}{5} + 1 \right)} \\ &= \frac{1}{5} \int \frac{dx}{\left(\frac{x}{\sqrt{5}} \right)^2 + 1} \\ &t = \frac{x}{\sqrt{5}} \Rightarrow dx = \sqrt{5} dt \\ &\Rightarrow \frac{\sqrt{5}}{5} \int \frac{1}{t^2 + 1} \\ &= \frac{\sqrt{5}}{5} \operatorname{Bgtan}(t) + C \\ &= \frac{\sqrt{5}}{5} \operatorname{Bgtan} \left(\frac{x}{\sqrt{5}} \right) + C \end{aligned}$$

References